

A predictive model for rapid assessment of protective efficacy against emerging SARS-CoV-2 variants

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ABSTRACT

An effective predictive model of protection would be very helpful to provide a timely and reliable evaluation of the vaccine induced protection against corresponding to rapidly emerging evolving SARS-CoV-2 variants. By integrating the validated “neutralizing antibody-vaccine efficacy” and “modified genetic distance-vaccine efficacy” models, we developed a composite sieve analysis framework (the “neutralizing antibody-genetic distance-vaccine efficacy” model) to predict the protective efficacy of COVID-19 vaccine regimens, particularly for different heterologous prime-boost COVID-19 vaccination regimens. Data for the model building were extracted from 23 published studies. Leave-one-out method was used to validate the model. Model validation demonstrated that the composite framework achieved high predictive accuracy, with concordance correlation coefficients of 0.95 (95% CI: 0.82–0.98) for the “neutralizing antibody-vaccine efficacy” submodel and 0.93 (95% CI: 0.49–0.99) for the “modified genetic distance-vaccine efficacy” submodel. Most prediction errors were within 5% and 10%, respectively. By applying this framework with neutralizing antibody data and SARS-CoV-2 variant sequencing data, we predicted the protective efficacy of different heterologous prime-boost regimens. The regimen of two-dose CoronaVac plus one-dose aerosolized Ad5-nCoV was estimated to confer 95.40% (95% CI: 92.67–98.13%) protection against symptomatic infection with wild-type SARS-CoV-2 at day 28 post-boost, 79.56% (95% CI: 53.31–100.00%) against the Delta variant, and 68.21% (95% CI: 42.53–93.89%) against Omicron BA.5.2.20. Predicted efficacy against other Omicron sublineages was generally below 50%, with near-zero efficacy for KP.2, KP.3 and XD.V.1. Compared with this regimen, two-dose CoronaVac plus one-dose intramuscular Ad5-nCoV booster or three-dose CoronaVac yielded consistently lower predicted efficacy across all variants. This study offers a generalizable approach for rapidly evaluating the efficacy of COVID-19 vaccines against emerging variants, providing timely evidence to guide vaccine deployment in future outbreaks.

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1. Introduction

The COVID-19 pandemic represents a severe global public health challenge. Vaccination is a cornerstone of control, yet the efficacy of primary regimens wanes over time, exacerbated by immune-evasive variants like Omicron (Andrews et al., 2022; Collie et al., 2022; Nordström et al., 2022). While previous study has showed that heterologous prime-boost strategies with different COVID-19 vaccines enhanced neutralizing antibodies and broaden cross-reactivity (Jin, Guo, et al., 2023; Jin, Tang, et al., 2023; Li, Hou, et al., 2022; Li, Wu et al., 2022). Although an increasing immune response associated with heterologous prime-boost regimen has been demonstrated, their protective superiority is inherently hampered by the rapid evolution and immune escape of SARS-CoV-2 variants. Conventional clinical trials often lag behind viral evolution, failing to provide timely efficacy data against circulating strains (Carabelli et al., 2023). In addition, efficacy trials of vaccines are costly, time-consuming, and ethical restrictions (Black, 2015). In the context of rapidly evolving viruses and emerging variants, relying solely on clinical trial study is insufficient for timely response to a changing pandemic landscape.

This critical lag necessitates alternative approaches. Neutralizing antibodies, strongly correlated with protection against infection across vaccine platforms and endorsed as correlates of protection (CoPs) by regulatory agencies, offer a key immunological metric (Earle et al., 2021; Gilbert et al., 2022; Regev-Yochay et al., 2023). Evidence from natural infection cohorts has demonstrated a strong positive correlation between neutralizing antibody titers and protection against reinfection (Addetia et al., 2020), with high antibody levels effectively preventing reinfection by circulating variants of concern (Garcia-Beltran et al., 2021). Consequently, neutralizing antibodies have been endorsed by both the U.S. Food and Drug Administration and the European Medicines Agency as immunological correlates of protection in support of emergency use authorization for COVID-19 vaccines (EMA, 2021; FDA, 2022). Khoury et al. integrated neutralizing antibody data from COVID-19 vaccines to reveal a nonlinear relationship between neutralizing antibodies and vaccine efficacy. The model relies on neutralizing antibody levels against specific strains post-vaccination. When new variants dominated, neutralizing antibody data for the new strain was required to predict efficacy (Khoury et al., 2021).

Genetic distance quantifies the sequence divergence between viral variants, reflecting key antigenic site mutations and immune escape potential, and serves as a critical metric for evaluating vaccine-induced immunity (Cerqueira-Silva, de Araujo Oliveira, et al., 2022; Cerqueira-Silva, de Araujo Oliveira, et al., 2022; Ella et al., 2021; Suphanchaimat et al., 2022). Therefore, the “genetic distance between viral variants and quantifying antigenic site divergence and immune escape potential” is a principal determinant of vaccine effectiveness, linking viral evolution to immune protection.

In this study, we developed a predictive model linking neutralizing antibody levels, genetic distance, and vaccine protection in order to address the challenge of timely efficacy assessment amidst evolving variants. Utilizing neutralization data from previous heterologous booster trials and introducing a refined genetic distance calculation focused on key RBD residues, our model enables rapid evaluation of heterologous booster efficacy against emerging SARS-CoV-2 variants.

2. Methods

2.1. Study design

Neutralizing antibody data from three previously reported heterologous prime-boost clinical trials of COVID-19 vaccines (ClinicalTrials.gov registration numbers: NCT04892459 (Li, Wu, et al., 2022), NCT04952727 (Jin, Tang, et al., 2023), and NCT05043259 (Li, Wu, et al., 2022); Supplementary Appendix 1) were used to construct the exponential decay models of antibody kinetics, which were subsequently applied to vaccine efficacy prediction. The three trial protocols were approved by Research Ethics Committee of Jiangsu Provincial Center for Disease Control and Prevention (JSJK2021-A007-01, JSJK2021-A008-01 and JSJK2021-A024-02), and were conducted following the principles of the Declaration of Helsinki, ICH Good Clinical Practice guidelines and local guidelines. Written informed consent was obtained from all participants. We established a “neutralizing antibody-vaccine efficacy” (NAb-VE) model based on two components: (1) published vaccine efficacy or effectiveness data from COVID-19 priming regimens and heterologous prime-boost regimens; and (2) published live-virus neutralization titres from the corresponding regimens. This model enabled prediction of both the magnitude and durability of protection conferred by different prime-boost vaccination strategies against SARS-CoV-2 variants (i.e. wild-type SARS-CoV-2, Delta, Omicron BA.1) for which neutralizing antibody titres were available.

Then, in order to extend the model predictions to other variants for which neutralizing antibody titres were unavailable, we established a second efficacy prediction model—the “modified genetic distance-vaccine efficacy” (GD-VE) model based on a combination of three components: (1) published vaccine efficacy or effectiveness data from heterologous COVID-19 prime-boost regimens; (2) predicted efficacy estimates derived from the NAb-VE model; and (3) viral genomic sequence data from the Global Initiative on Sharing All Influenza Data (GISAID) database.

At last, we developed a sieve analysis framework—the “neutralizing antibody-genetic distance-vaccine efficacy” (NAb-GD-VE) model to predict average vaccine efficacy against symptomatic COVID-19 diseases by integrating the NAb-VE and GD-VE models. Using monthly whole-genome sequencing data during the pandemic and post-pandemic periods (July 2021–July 2023) in Jiangsu Province, China, we estimated temporal changes in efficacy across different vaccination strategies as SARS-CoV-2 variants circulated. This prediction assumed that, for months without reported COVID-19 cases, the genetic distance was taken as identical to that of the preceding month with circulating variants to ensure continuity in monthly estimates. The study evaluated three vaccination regimens: (1) two doses of inactivated CoronaVac followed by an aerosolized Ad5-nCoV booster; (2) two doses of CoronaVac followed by an intramuscular Ad5-nCoV booster; and (3) three doses of homologous CoronaVac.

2.2. Data extraction

In order to establish NAb-VE model and GD-VE model, we systematically searched PubMed and Web of Science for studies on COVID-19 vaccine immunogenicity and efficacy or effectiveness ([Supplementary Appendix 2](#)).

A total of 23 studies were included to establish the models ([Fig. S3](#)). The NAb-VE model was fitted using data from 9 studies reporting live-virus neutralizing antibody titres standardized to convalescent sera (available either directly or indirectly) induced by COVID-19 vaccines ([Angkasekwinai et al., 2024](#); [Folegatti et al., 2020](#); [Jackson et al., 2020](#); [Keech et al., 2020](#); [Logunov et al., 2020](#); [Sadoff et al., 2021](#); [Walsh et al., 2020](#); [Wheatley et al., 2021](#); [Zhang et al., 2021](#)), as well as 9 studies reporting vaccine efficacy or effectiveness against symptomatic SARS-CoV-2 infection ([Baden et al., 2021](#); [Jara et al., 2022](#); [Logunov et al., 2021](#); [Lumley et al., 2021](#); [Novavax, 2021](#); [Pearson, Magalhaes, & Deng, 2021](#); [Polack et al., 2020](#); [Sacks, 2021](#); [Voysey et al., 2021](#)).

For the GD-VE model, to enable subsequent application in predicting the efficacy of heterologous boosters after inactivated vaccine priming, we combined two data sources: (i) efficacy estimates derived from the NAb-VE model for three regimens—two doses of CoronaVac followed by aerosolized Ad5-nCoV booster, two doses of CoronaVac followed by intramuscular Ad5-nCoV booster, and three doses of homologous CoronaVac; and (ii) observed effectiveness data from five studies of heterologous prime-boost regimens: two doses of CoronaVac followed by BNT162b2 (reported in all five studies ([Cerqueira-Silva, de Araujo Oliveira, et al., 2022](#); [Cerqueira-Silva, de Araujo Oliveira, et al., 2022](#); [Jara et al., 2022](#); [Sritipsukho et al., 2022](#); [Suphanchaimat et al., 2022](#))) and two doses of CoronaVac followed by ChAdOx1 (reported in three studies ([Jara et al., 2022](#); [Sritipsukho et al., 2022](#); [Suphanchaimat et al., 2022](#))). A total of 87,036 full-length SARS-CoV-2 genome sequences (FASTA format) corresponding to the time and location of the five studies were downloaded from the GISAID EpiCoV database (<https://gisaid.org/>) and were used to compute genetic distances for the GD-VE model construction ([Supplementary Appendix 3](#)). In addition, 1701 full-length genome sequences were collected from Jiangsu Province, China, between July 2021 and July 2023, were downloaded from the GISAID EpiCoV database. These sequences were used to calculate genetic distances and to apply the NAb-GD-VE sieve model for predicting the dynamic vaccine efficacy during periods of overlapping variant circulation.

2.3. Statistical methods

2.3.1. Modeling for vaccine efficacy based on neutralizing antibody levels

Following the Prentice Criteria outlined in the WHO report on evaluating immunological surrogate endpoints for vaccine protection ([Supplementary Appendix 4](#)) ([WHO, 2013](#)), we assessed whether neutralizing antibody titres against live SARS-CoV-2 virus could serve as an immunological surrogate endpoint for protection conferred by COVID-19 vaccination regimens.

The model used a logistic relationship to link neutralizing antibody levels to protection probability and applied maximum likelihood estimation to fit the model and determine optimal parameters for population-level vaccine efficacy. This model was based on the fundamental assumptions: (i) the vaccine efficacy was primarily mediated by neutralizing antibodies; (ii) live-virus neutralizing antibodies declined exponentially over time; and (iii) their levels were linked to efficacy through a logistic function ([Supplementary Appendix 5](#)).

The model was internally validated using leave-one-out cross-validation and externally validated with observed vaccine efficacy data from heterologous prime-boost regimens not included in the model. Sensitivity analysis was conducted using the 95% CI bounds of the mean neutralizing antibody levels to predict efficacy ([Supplementary Appendix 5.4](#)). Agreement between model-predicted and observed efficacy values was assessed using the Concordance Correlation Coefficient (CCC). This modeling framework was originally proposed by Khoury et al. ([Khoury et al., 2021](#)), and we further refined the model by incorporating heterologous prime-boost vaccination data.

We utilized live virus neutralizing antibody data from three previously conducted clinical trials on heterologous prime-boost COVID-19 vaccination regimens ([ClinicalTrials.gov](#) registration numbers: NCT04892459 ([Li, Hou, et al., 2022](#)), NCT04952727 ([Jin, Guo, et al., 2023](#)), and NCT05043259 ([Li, Wu, et al., 2022](#))) to fit an exponential decay model ([Supplementary Appendix 5.5](#)). This was then incorporated into the NAb-VE model to estimate the durability of vaccine-induced protection.

2.3.2. Modeling for vaccine efficacy based on genetic distance

The genetic distance-based prediction model was constructed using a two-level linear mixed-effects model (Cao et al., 2022). This model relies on three assumptions: (i) immune escape of SARS-CoV-2 variants after heterologous prime-boost regimens was primarily determined by genetic distance; (ii) vaccine efficacy declines linearly with increasing genetic distance; (iii) and covariates such as age and time since booster acted independently of genetic distance (Supplementary Appendix 6).

Previous studies have commonly used the Hamming distance to calculate genetic distance between viral sequences, assigning a uniform weight of 1.0 to each amino acid mismatch and computing the average number of mismatches (Cao et al., 2022). In the present study, we accounted for the fact that key mutations tend to accumulate at antigenic sites within the receptor-binding domain (RBD) of SARS-CoV-2 variants—mutations that substantially alter antigenicity and facilitate escape from neutralizing antibodies (NAbs) elicited by repeated vaccination or infection. A total of 13 critical amino acid positions were selected as follows: R346, K378, K417, K444, V445, G446, N450, L452, V483, E484, F486, F490, and Q493 (Jian et al., 2025). These 13 sites repeatedly mutate across multiple SARS-CoV-2 variant lineages, enhancing viral infectivity and increasing escape from monoclonal antibody capture (Jian et al., 2025). To reflect the increased immunological relevance of these sites, we assigned greater weights to mismatches occurring at these critical antigenic positions when calculating the average number of amino acid mismatches, thereby defining a modified genetic distance. Based on the deep mutational scanning (DMS) scores of these key mutations reflecting immune escape *in vitro*, which differed by up to twofold, we experimentally set the weight coefficients for key antigenic sites to 1.0, 1.5, and 2.0, as no related model parameters have been reported previously (Dadonaité et al., 2023; Jian et al., 2025). Using these weights, we recalculated the modified genetic distances and subsequently fitted each to the GD-VE model (Supplementary Appendix 6.2). The performance of each model was evaluated by calculating the CCC between the predicted and validated vaccine efficacy values. The weight yielding the highest CCC was selected as optimal.

The GD-VE model was validated similarly to the NAb-VE model, using leave-one-out cross-validation, external data, sensitivity analysis, and agreement assessed with the CCC (Supplementary Appendix 6.4).

2.3.3. Sieve analysis model for predicting vaccine efficacy

We developed a composite sieve analysis framework of the NAb-GD-VE model using a weighted averaging approach under conditions of overlapping and sequential circulation of diverse SARS-CoV-2 variants (Supplementary Appendix 7). As this model represents an integration of the validated NAb-VE and GD-VE models, no separate validation was performed.

Molecular phylogenetic analysis of full-length SARS-CoV-2 genomes was performed to explore the relationship between predicted vaccine efficacy and genetic divergence of variants (Supplementary Appendix 8).

All statistical analyses and data visualizations were performed using R software (version 4.2.3). When the predicted efficacy boundary exceeds the predefined bounds of 0.00% and 100.00%, a truncation method has been applied to ensure statistical validity. The primary R packages used were listed in Table S4. The cloud-based online prediction system was developed using Python (version 3.0) with the Flask framework (see Supplementary Appendix 10 for system development details).

3. Results

3.1. “Neutralizing antibody-vaccine efficacy” model fitting and validation

The results showed that neutralizing antibodies can thus serve as valid immunological surrogate endpoints for COVID-19 vaccine protection satisfying the Prentice criteria (Supplementary Appendix 11). In this study, the fitted parameter values for the NAb-VE model were $\sigma = 0.4550$, $k = 2.2924$ (95% CI: 1.8612–2.8234), and $n_{50} = 0.1784$ (95% CI: 0.1192–0.2669) (Fig. 1a).

As the fold decrease in neutralizing antibodies against wild-type SARS-CoV-2 increases, predicted efficacy against symptomatic infection declined for all three regimens: two-dose inactivated CoronaVac plus an aerosolized Ad5-nCoV, two-dose CoronaVac plus one-dose intramuscular Ad5-nCoV, and three-dose CoronaVac, with a more gradual decline for those showing higher protection against the wild-type strain (Fig. 1b).

The internal and external validation results showed close agreement between predicted and observed efficacy (Fig. 1c and d). The error was almost always within 5% (Table S8 and Table S9). The calibration plot showed strong concordance between predicted and observed efficacy, with a CCC of 0.95 (95% CI: 0.82–0.98) (Fig. 1e). Sensitivity analysis confirmed the robustness of the model predictions (Fig. 1f and Table S10).

3.2. “Modified genetic distance-vaccine efficacy” model fitting and validation

The GD-VE model results showed that genetic distances in various SARS-CoV-2 genomic regions, including RBD, were significantly linked to protection efficacy against symptomatic COVID-19 ($P < 0.05$). RBD genetic distance, influencing viral binding to ACE2, was selected, with each RBD mutation associated with a 2.27% reduction in protection (Table S11).

To capture the immunological significance of RBD antigenic sites, we applied weighted schemes to key positions and recalculated genetic distances for GD-VE model fitting. A weight coefficient of 2.0 yielded the highest CCC in validation and

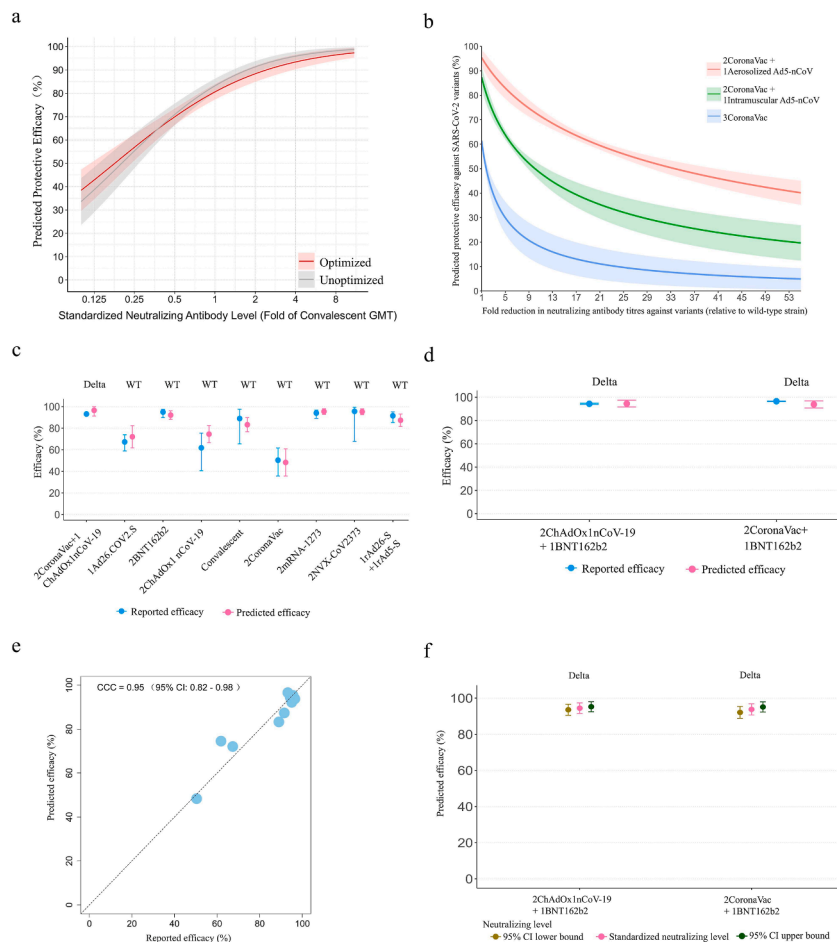


Fig. 1. Modeling prediction on the efficacy against symptomatic SARS-CoV-2 infection based on neutralizing antibody titres after heterologous prime-boost immunization.

(a) Evaluation curves of predicted protective efficacy against symptomatic SARS-CoV-2 infection based on live virus neutralizing antibody titres. The red curve represents the predictive model established in this study, in which neutralizing antibodies were used to estimate vaccine-induced protective efficacy following heterologous prime-boost immunization. The model parameters were $\sigma = 0.4550$, $k = 2.2924$, and $n_{50} = 0.1784$. The grey curve denotes the model developed by Prof. Khoury and colleagues to predict protective efficacy following primary immunization ($\sigma = 0.4647$, $k = 3.0977$, $n_{50} = 0.2011$). Shaded regions indicate 95% CIs, calculated as predicted values ± 1.96 standard errors, with standard errors estimated from the Hessian matrix of the likelihood function. The red prediction curve has a slightly smaller slope compared to the pre-optimization grey curve, making it more gradual. This suggests that the protection efficacy against symptomatic infections increases more gradually with rising neutralizing antibody levels. The slight decrease in n_{50} and the slight leftward shift of the curve indicate a slight reduction in the neutralizing antibody level required for 50% protection. (b) Predicted protective efficacy against symptomatic infection with SARS-CoV-2 variants under different heterologous prime-boost immunization strategies. Predictions are based on the established “neutralizing antibody–vaccine efficacy” model, in which efficacy is modelled as a function of the fold reduction in neutralizing antibody titres against SARS-CoV-2 variants relative to wild-type SARS-CoV-2. Curves of different colors represent distinct heterologous prime-boost immunization regimens and their predicted protective efficacy against symptomatic infection across a range of neutralization fold-reductions. Shaded areas indicate 95% CIs. (c) Internal validation of the “neutralizing antibody–vaccine efficacy” model predicting efficacy against symptomatic SARS-CoV-2 infection. Each predicted efficacy value was generated using a leave-one-out cross-validation approach. Different colored shapes indicate predicted and reported efficacy values. Circles denote point estimates of efficacy, and error bars represent 95% CIs. (d) External validation of the “neutralizing antibody–vaccine efficacy” model predicting efficacy against symptomatic SARS-CoV-2 infection. The heterologous prime-boost regimen of two doses of CoronaVac followed by one dose of BNT162b2 was excluded from model fitting due to signs of over-fitting and poor performance during internal validation, and was thus retained only for external validation. Two doses of ChAdOx1 nCoV-19 followed by one dose of BNT162b2 was also for external validation. Circles denote point estimates of efficacy, and error bars represent 95% CIs. (e) Calibration plot for internal and external validation of the “neutralizing antibody–vaccine efficacy” model predicting efficacy of heterologous prime-boost immunization against symptomatic SARS-CoV-2 infection. CCC, concordance correlation coefficient. (f) Sensitivity analysis of the “neutralizing antibody–vaccine efficacy” model predicting efficacy of heterologous prime-boost immunization against symptomatic SARS-CoV-2 infection. Sensitivity analysis was performed by predicting efficacy using the standardized neutralizing antibody level as well as its upper and lower bounds of the 95% CI. Circles indicate predicted efficacy values, with error bars denoting their corresponding 95% CIs. CI, confidence interval.

was adopted as the optimal weighting strategy (Table S12). With the wild-type SARS-CoV-2 strain as the reference, modified RBD genetic distances for variants ranged from 3 for Delta to 37 for XDV.1 (Table S13).

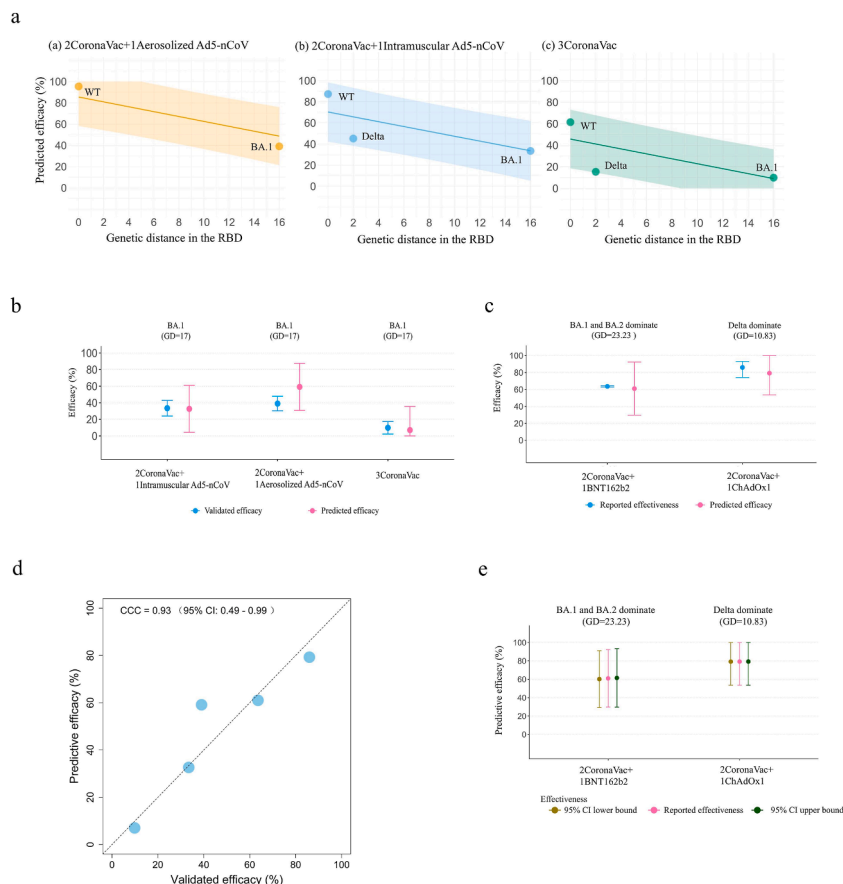


Fig. 2. Modeling prediction on the efficacy against symptomatic SARS-CoV-2 infection based on modified RBD genetic distance.

(a) Predicted efficacy against symptomatic SARS-CoV-2 infection based on modified RBD genetic distance across heterologous prime-boost immunization regimens. Solid lines represent the predicted relationship between efficacy and the modified genetic distance of the receptor-binding domain (RBD) of SARS-CoV-2 for different heterologous prime-boost immunization regimens. Shaded areas indicate 95% CIs. Circular dots denote validated efficacy values, predicted using the “neutralizing antibody–vaccine efficacy” model. (b) Internal validation of the “modified genetic distance–vaccine efficacy” model predicting efficacy against symptomatic SARS-CoV-2 infection. Both the validated and predicted efficacy values represent efficacy against symptomatic SARS-CoV-2 infection at a specific modified viral genetic distance. (c) External validation of the “modified genetic distance–vaccine efficacy” model. Reported and predicted efficacies refer to protection against symptomatic infection caused by variants with specific modified genetic distances. Dots and error bars represent point estimates and 95% confidence intervals of predictive efficacy. (d) Calibration plot for internal and external validation of the “modified genetic distance–vaccine efficacy” model predicting efficacy of heterologous prime-boost immunization against symptomatic SARS-CoV-2 infection. CCC, concordance correlation coefficient; CI, confidence interval. (e) Sensitivity analysis of the “modified genetic distance–vaccine efficacy” model of heterologous booster regimens. Sensitivity analysis was performed by fitting the model using the reported efficacy, as well as the upper and lower bounds of the 95% CI, corresponding to the highest modified genetic distance from one heterologous prime-boost regimen used in external validation. Dots indicate predicted efficacy values, and error bars represent their 95% CIs. CI, confidence interval. GD, modified genetic distance.

The GD-VE model achieved an R^2 of 85.53% (Table S15) with intercept and error term standard deviations of 28.26 and 13.33, respectively (Table S17). The model's predicted relationship was shown in Fig. 2a. Regression diagnostics indicated a good fit of the GD-VE model (Figure S6–S8).

Internal and external validation showed that across all heterologous prime-boost regimens, predicted and observed efficacy differed by <10%, with overlapping 95% CIs (Fig. 2b and Table S20; Fig. 2c and Table S23). The CCC between the predicted and observed efficacy values in internal and external validation was 0.93 (95% CI: 0.49–0.99), indicating strong agreement between predicted and actual outcomes (Fig. 2d). Sensitivity analyses further confirmed the robustness of the GD-VE model (Fig. 2e and Table S24).

3.3. Efficacy of heterologous prime-boost COVID-19 vaccination regimens

The validated NAb-VE and GD-VE models predicted that, among the three heterologous prime-boost regimens, two doses of CoronaVac followed by an aerosolized Ad5-nCoV booster conferred the highest predictive efficacy against symptomatic COVID-19, followed by the intramuscular Ad5-nCoV booster, whereas the homologous three-dose CoronaVac regimen provided the lowest predictive protection. For the regimen of two doses of CoronaVac followed by an aerosolized Ad5-nCoV,

predicted efficacy was 95.40% (95% CI: 92.67%–98.13%) against the wild-type strain, 79.56% (95% CI: 53.31%–100.00%) against Delta, and 68.21% (95% CI: 42.53%–93.89%) against Omicron BA.5.2.20. Against other Omicron sublineages (BA.2.75, BA.1, BF.5 and BF.7), predicted efficacy fell below 50%, and for EG.1.2, XBB.1.16, EG.5.1, JN.1, XDV.1, KP.2, and KP.3, the model predicted little to no protection.

Predicted efficacy was 87.32% (95% CI: 83.86–90.79) for two doses of CoronaVac followed by an intramuscular Ad5-nCoV booster and 61.40% (56.64–66.17) for three doses of CoronaVac against the wild-type strain. However, for nearly all other variants, both regimens exhibited <50% predicted efficacy or no measurable protection estimates (Fig. 3a and Table S25).

A phylogenetic tree based on whole-genome sequences of 14 SARS-CoV-2 strains and 29,743 nucleotide positions showed that closer evolutionary relationships corresponded to more similar predicted protective efficacy against symptomatic infection under heterologous prime-boost regimens (Fig. S9).

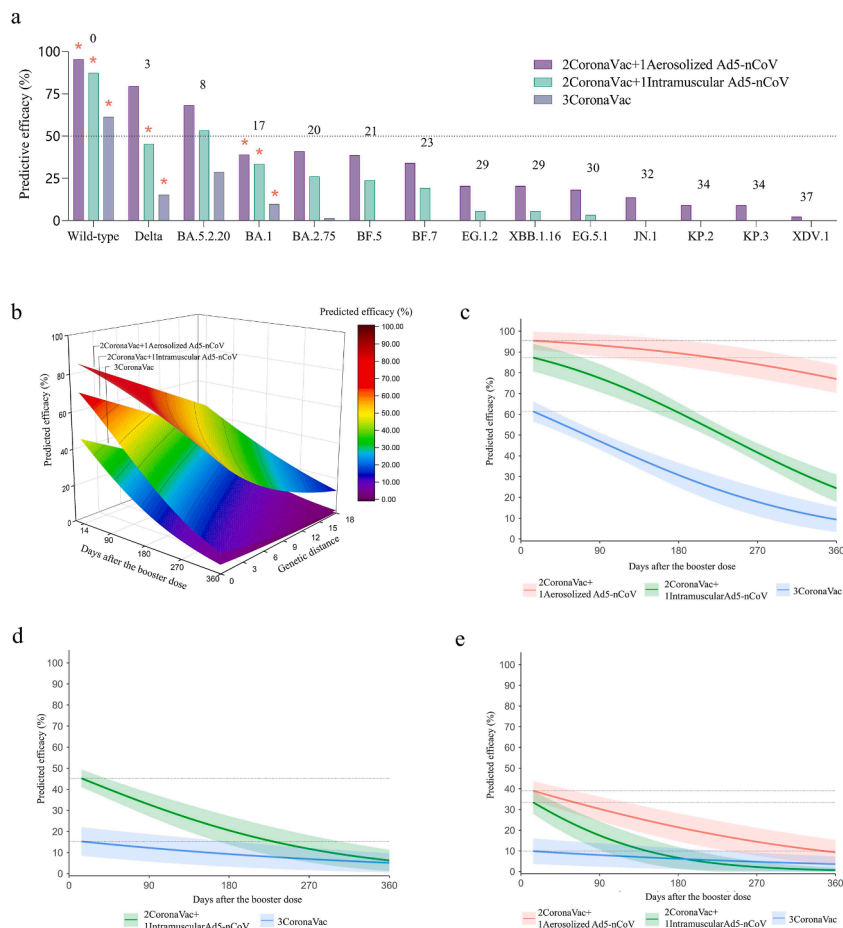


Fig. 3. Predicted efficacy and durability of protection against symptomatic infection with the different SARS-CoV-2 strain.

(a) Predicted efficacy of different heterologous prime-boost regimens against symptomatic infection with specific SARS-CoV-2 variants. Orange asterisks (*) indicate efficacy predictions based on "neutralizing antibody-vaccine efficacy" model and all other predictions are derived from the "modified genetic distance-vaccine efficacy" model. The horizontal dashed line represents 50% predictive efficacy. Numbers above each bar represent the modified genetic distance of each variant from the wild-type SARS-CoV-2 strain. (b) Predicted vaccine efficacy over time and modified genetic distance for different heterologous prime-boost regimens. Three-dimensional surface plots illustrate the predicted protective efficacy (%) over time (days after booster dose, x-axis) and across viral variants with increasing modified genetic distance (z-axis) for three immunization regimens. The color gradient reflects the level of predicted efficacy, with warmer colors indicating higher efficacy and cooler colors indicating lower efficacy. Each surface demonstrates the waning of efficacy over time and across genetic divergence, highlighting the differences in durability and breadth of protection conferred by each booster regimen. (c) Durability of protection against symptomatic infection with the wild-type SARS-CoV-2 strain under different heterologous prime-boost regimens. Shaded areas represent 95% CIs. Horizontal dashed lines indicate the initial predicted protection efficacies: 95.40% for two doses of CoronaVac followed by aerosolized Ad5-nCoV, and 61.40% for three doses of CoronaVac. (d) Durability of protection against symptomatic infection with the Delta variant under different heterologous prime-boost regimens. Shaded areas represent 95% CIs. The horizontal dashed lines indicate the initial predicted protection efficacies against Delta: 45.23% for two doses of CoronaVac followed by intramuscular Ad5-nCoV, and 15.28% for three doses of CoronaVac. (e) Durability of protection against symptomatic infection with the Omicron BA.1 variant under different heterologous prime-boost regimens. Shaded areas represent 95% CIs. The horizontal dashed lines indicate the initial predicted protection efficacies: 39.03% for two doses of CoronaVac followed by aerosolized Ad5-nCoV, 33.44% for two doses of CoronaVac followed by intramuscular Ad5-nCoV, and 9.87% for three doses of CoronaVac. CI: confidence interval.

3.4. Vaccine efficacy in serial cross-sectional sequencing data

Between July 2021 and July 2023 in Jiangsu, China, multiple SARS-CoV-2 variants cross-circulated. In this context, sieve model analyses showed that two doses of CoronaVac followed by an aerosolized Ad5-nCoV booster consistently provided the highest predicted efficacy against symptomatic COVID-19. However, after Omicron sublineages emerged, the predicted efficacy of all three regimens dropped below 50% (Fig. 4).

3.5. Prediction of the potential for long-term protection

Neutralizing antibody data obtained from clinical trials were used to fit exponential decay models for each SARS-CoV-2 strain, with estimated half-lives summarized in Table S26–S28.

The GD-VE model predicted that two doses of CoronaVac followed by an aerosolized Ad5-nCoV booster provided the most durable estimated protection against symptomatic infection with the wild-type strain, with efficacy of 76.98% (95% CI: 70.32%–83.63%) at 12 months and 36.76% (95% CI: 31.71%–41.81%) at 24 months post-boost. Efficacy at 12 months for the two doses of CoronaVac followed by an intramuscular Ad5-nCoV and three-dose CoronaVac regimens was 24.39% (95% CI: 17.85%–30.94%) and 9.33% (95% CI: 3.34%–15.33%), respectively, both declining to nearly zero by 24 months post-boost (Fig. 3c and Table S29).

For the Delta variant, two doses of CoronaVac followed by an intramuscular Ad5-nCoV regimen yielded predicted efficacies below 50% throughout, while the three-dose CoronaVac regimen remained consistently below 15% (Fig. 3d and Table S30). Durability efficacy estimates for the two doses of CoronaVac followed by an aerosolized Ad5-nCoV regimen were not analyzed for Delta, as neutralizing antibody measurements for this variant were not pre-specified in the protocol.

For the Omicron BA.1 variant, both the two doses of CoronaVac followed by an aerosolized or intramuscular Ad5-nCoV regimens conferred relatively low levels of predicted protection, while the homologous three-dose CoronaVac regimen provided virtually none (Fig. 3e and Table S31).

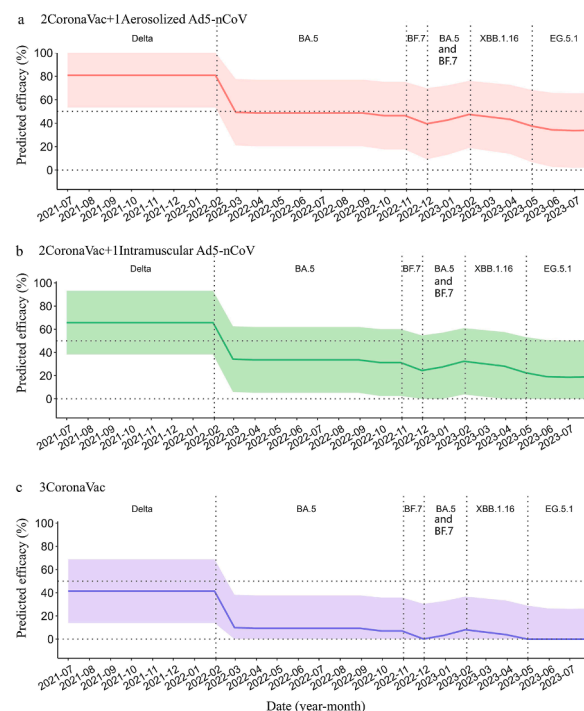


Fig. 4. Predicted protection efficacy against symptomatic SARS-CoV-2 infection of different heterologous prime-boost regimens based on viral sequencing data from Jiangsu Province, China.

Variant names correspond to the predominant circulating strains during each time period, in the context of co-circulation of multiple variants. The upper horizontal dashed line marks 50% predictive efficacy, and the lower line marks 0%. Vertical dashed lines separate distinct variant-dominant periods, labeled above with the respective leading strains. Solid lines represent predicted protection efficacy, and shaded areas denote the 95% CIs. Predictions were performed using a sieve model based on neutralizing antibody titers and modified genetic distance, informed by Jiangsu Provincial SARS-CoV-2 sequencing data. Following the emergence of Omicron sublineages, predicted efficacies against symptomatic infection were below 50% across all heterologous regimens. Of the three prime-boost regimens, two doses of CoronaVac followed by aerosolized Ad5-nCoV yielded the highest predicted protection. CI: confidence interval.

We further assumed that the half-life of neutralizing antibodies against variants with different genetic distances was the same as that against the wild-type strain, enabling prediction of vaccine efficacy over time and genetic distance. Predicted efficacy declined with both time and increasing genetic distance across all three regimens (Fig. 3b).

Additionally, we developed a cloud-based online prediction platform for the NAb-VE model, accessible at: <http://8.153.70.192:1003/>. The platform requires no login, and international users can directly perform efficacy predictions. The system functionality and interface were shown in Fig. S10. We demonstrated the utility of the platform using two case examples, which confirmed its reliable predictive performance (Supplementary Appendix 20).

4. Discussion

The ongoing evolution of SARS-CoV-2 and the risk of future respiratory pandemics highlight the need for efficient tools to evaluate vaccine-induced protection (Bardhan et al., 2024; WHO, 2024). This study established a predictive framework that integrates neutralizing antibody titers and viral genetic distance to estimate the efficacy and durability of heterologous prime-boost immunization. This model enables real-time estimation of protection against symptomatic infection caused by specific SARS-CoV-2 variants, quantifies the durability of efficacy across heterologous prime-boost regimens, and evaluates average efficacy under overlapping variant circulation. The framework provides a scalable approach to inform vaccine deployment in future outbreaks.

In developing the neutralizing antibody-based protection model, we built upon the framework established by Khoury et al. (Khoury et al., 2021), optimizing model parameters using data from heterologous prime-boost immunization studies and adjusting for covariates such as pre-existing Ad5 immunity, which may impair the generation of protective antibodies induced by Ad5-nCoV (Li et al., 2023; Tang et al., 2023). Our optimized model better reflects the characteristics of heterologous boosting with adenoviral vector following inactivated vaccine priming.

In this study, we developed a genetic distance-based protection prediction model to estimate the efficacy of different heterologous prime-boost regimens against emerging variants with unknown neutralizing antibody levels, thereby addressing a limitation of the neutralization-based models. We found that amino acid substitutions in the RBD significantly affected the predicted efficacy of heterologous regimen against symptomatic SARS-CoV-2 infection—consistent with previous findings in the context of primary vaccination (Cao et al., 2022). The RBD is the principal target of neutralizing antibodies, and most COVID-19 vaccines (e.g., mRNA and inactivated vaccines) confer protection primarily by inducing RBD-specific neutralizing antibodies (Liang et al., 2025; Liu et al., 2023). Mutations in the RBD can alter antibody-binding sites, reducing the binding affinity of neutralizing antibodies and thereby diminishing vaccine-induced protection (Ao et al., 2022). SARS-CoV-2 variants continue to accumulate recurrent mutations at key antigenic sites within the RBD, leading to increased infectivity and enhanced escape from both monoclonal antibodies and vaccine or infection-induced neutralizing antibodies (Jian et al., 2023, 2025; Kaku et al., 2023; Yang et al., 2024). Our framework is the first to define a modified RBD genetic distance that assigns greater weight to key antigenic sites, enhancing the metric's ability to capture biologically meaningful variation relevant to immune escape and vaccine performance.

This study establishes a cloud-based prediction system for rapid estimation of vaccine efficacy (with 95% CIs) against specific SARS-CoV-2 variants via an intuitive web interface. The platform delivers real-time visualization, stable performance, and requires no statistical expertise. Its generalizable framework allows adaptation to other respiratory pathogens following parameter validation, serving as a scalable solution for vaccine evaluation and pandemic preparedness. By leveraging existing viral surveillance infrastructure, this system enables real-time efficacy assessment during multi-variant circulation and provides a technical foundation for evaluating vaccines against future emerging respiratory threats.

This study has several limitations. Firstly, WHO reference standards may cause slight underestimation of predicted efficacy, though this effect attenuates at higher antibody levels. Secondly, the neutralization models used clinical trial efficacy data while genetic distance models incorporated real-world effectiveness data. Thirdly, this study is limited by its focus on heterologous sequential booster regimens following inactivated COVID-19 vaccines, primarily due to data availability. Expanding the external validation dataset to include additional vaccine types and diverse population characteristics would be essential for improving the model's generalizability and prediction accuracy in various scenarios. This limitation should be addressed in future research. Finally, a significant limitation of our model is the exclusive focus on humoral immunity, which does not account for the roles of cellular immunity (e.g., T-cell responses) and mucosal immunity. Both cellular and mucosal immunity play crucial roles in the protection against SARS-CoV-2 infection, particularly in preventing severe disease and influencing viral clearance. Excluding these immune responses could lead to an incomplete understanding of vaccine efficacy, especially against variants with immune escape capabilities. Future models should integrate data on T-cell responses, mucosal immunity, and their interactions with neutralizing antibodies to provide a more comprehensive evaluation of vaccine-induced protection. This would help reduce potential biases and enhance the accuracy and applicability of the prediction.

5. Conclusions

This study developed a framework for evaluating the efficacy of heterologous prime-boost COVID-19 vaccination based on neutralizing antibodies and viral genetic distance. By leveraging existing viral surveillance networks, the framework

enables real-time estimation of vaccine efficacy under co-circulating variant scenarios and offers a technical foundation for timely evaluation of vaccine efficacy against future emerging respiratory pathogens.

CRedit authorship contribution statement

Lairun Jin: Writing – original draft, Methodology, Data curation. **Siyue Jia:** Methodology, Data curation. **Chengwei Shao:** Data curation. **Ruifan Shen:** Data curation. **Pengfei Jin:** Writing – review & editing, Supervision, Conceptualization. **Jingxin Li:** Writing – review & editing, Supervision, Conceptualization.

Ethical approval statement

No human subjects were involved in this study, so ethical approvals were not required for this manuscript.

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Data availability

All data and code used to build the predictive models are publicly available at <https://github.com/18298261189/Rapid-Assessment-of-Protective-Efficacy>. Viral sequence data were downloaded from the GISAID at <http://platform.gisaid.org/>.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: None.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.idm.2026.03.003>.

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